

Introduction to Microeconomics –S102015CR

Prof. Federica Sbergami

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Problem Set 6

A. Externalities

Exercise 1

Suppose that the electricity market is in a situation of perfect competition. Demand takes the following form: $Q^D(P) = 7 - P$, and supply takes the following form: $Q^S(P) = P - 1$.

Suppose that the electricity produced in this market comes from coal-fired power plants that emit greenhouse gases that contribute to global warming and fine particles that can cause serious health problems. The government estimates that electricity production generates environmental and health costs of \$5 for each kW traded on this market (denoted q).

1. What is the socially optimal quantity of electricity traded on this market?
2. What is the amount of tax applied to electricity?

Exercise 2

Eating seasonal fruits and vegetables has a positive externality, given the reduced impact of this consumption mode on the environment.

The demand function for seasonal fruits and vegetables is given by $Q_d = 48 - P$ and the supply function is given by $Q^s = 16 + P$. The marginal social benefit of this good is $B_{ms} = 56 - Q$.

Which of the following statements is correct?

- (a) An optimal Pigouvian subsidy would increase the quantity consumed by 4 units.
- (b) An optimal Pigouvian subsidy would lower the price paid by consumers by 6 CHF.
- (c) An optimal Pigouvian subsidy would set the price paid by consumers at 16. (d) With an optimal Pigouvian subsidy, the quantity traded would be 32 units.

Exercise 3 (Exam January 2024)

The consumption of alcoholic beverages creates negative externalities because it affects the health of individuals and impacts the public health system. The demand function for alcoholic beverages is given by $Q_D = 36 - P$. The supply function is given by $Q_S = P$ and the marginal social cost of this good is $MSC = Q + 4$. The externality is therefore an additional external cost of CHF 4 per unit consumed. Which of the following statements is correct?

- (a) An optimal Pigouvian tax would reduce the quantity consumed by 2 units.
- (b) An optimal Pigouvian subsidy would lower the market price by CHF 6.

- (c) An optimal Pigouvian tax would set a market price of CHF 16.
- (d) An optimal Pigouvian subsidy would set a market price of CHF 16.

Exercise 4 (Exam January 2020)

The demand function for plane trips between Geneva and Barcelona is given by $Q^D(P) = 40 - P$. Its supply function is $Q^S(P) = P$ and the marginal social cost (including the private marginal cost) is $MSC = Q + 20$.

Which of the following propositions is correct?

- (a) An optimal environmental (Pigouvian) tax would reduce the amount of plane trips consumed by 10.
- (b) An optimal environmental (Pigouvian) subsidy would increase the amount of plane trips consumed by 10.
- (c) An optimal environmental (Pigouvian) tax would establish a market price of 15.
- (d) An optimal environmental (Pigouvian) subsidy would increase the amount consumed to 30.

B. Public goods Exercise 5 (Exam January 2021)

Among the following goods and services, which ones are simultaneously excludable and non-rival?

- (a) Solar energy.
- (b) Netflix and other video-on-demand services.
- (c) Polluting permits.
- (d) Deep sea fisheries

Exercise 6 (Exam January 2021)

Among the following statements, which one is correct?

- (a) Solar energy is a public good, but the electricity generated by solar panels is a private good.
- (b) The use of fossil fuels leads to a positive externality, which can be internalized by a Pigouvian subsidy.
- (c) If different firms face heterogeneous costs of reducing CO2 emissions, then the introduction of a Pigouvian tax on CO2 emissions leads to heterogeneous reductions in CO2 emission; the firms that depollute the most are those with the least elastic marginal cost of depollution.
- (d) Fossil fuels are characterized by exclusion in consumption, while solar energy is a resource whose consumption is characterized by rivalry.

Exercise 7 (Exam January 2020)

Four farmers are deciding whether to contribute to the maintenance of an irrigation project. For each farmer, the cost of contributing to the project is 20 CHF. For each farmer who contributes, all four of them benefit from an increase in their crop yields corresponding to a gain of 8 CHF each.

Which of the following statements is correct?

- (a) If all the farmers are selfish then none contribute.
- (b) If all the farmers are selfish then they all contribute 10 CHF.
- (c) The Pareto optimal outcome is for all farmers to contribute nothing.
- (d) If one of the farmers cares about one of her neighbours just as much as about herself then she contributes \$10.

C. Uncertainty Exercise 8 (Exam January 2013)

Consider an individual who is offered a lottery that pays 10CHF with a probability of 90% and 20CHF with a probability of 10%. We also know that this individual will always prefer to obtain 11CHF with certainty rather than play this lottery. Which of the following statements is **false**?

- (a) The individual is risk averse
- (b) This individual always accepts a fair bet
- (c) If the individual is able to determine a risk premium, it would be positive.
- (d) The individual's utility function is concave.

Exercise 9 (Exam January 2017)

Nicolas proposes the following game to his friend Pierre: he tosses a coin. If heads appear, the game ends and Pierre wins 2 CHF. Otherwise, he tosses the coin again. If heads appear on the second toss, Pierre wins 4 CHF. Otherwise, Nicolas tosses the coin again. If heads appear on the third toss, Pierre wins 8 CHF, otherwise, he wins nothing.

Which of the following propositions is necessarily correct?

- (a) If Peter is risk neutral, he is willing to pay up to 3 CHF to Nicholas to be able to participate in this game.
- (b) Only risk-averse players are willing to pay to be able to participate in this game.
- (c) Risk-averse players will never pay to participate in this game. (d) If Peter is risk-neutral, he is willing to pay up to 8 CHF to be able to participate in this game.

Exercise 10 (Exam January 2018)

It is 10 am and Louis wants to sell 10 Bitcoins that he owns. He can sell them now or wait until 4 pm. The current unit value of the Bitcoin is 10,000 CHF. At 4 pm, the value of each Bitcoin will be 15,000 CHF with a probability of 60%, 4,000 CHF with a probability of 20%, or 1,000 CHF with a probability of 20%.

Consider the following propositions:

1. Since there is a greater probability that the price will rise, Louis will wait until 4 pm, regardless of his preferences.
2. If Louis is risk averse, then he will sell his Bitcoins at 4 pm.
3. If Louis is risk neutral then he is indifferent between selling his Bitcoins at 10 am or 4 pm.

Which of the following statements is correct?

- (a) Only 3 is correct.
- (b) Only 1 is correct.
- (c) 1 and 2 are correct. (d) 2 and 3 are correct.

D. Additional exercises

Exercise 11 (Bonus exam)

Consider the fertilizer market with the inverse demand given by $P^D = 84 - Q$ and the supply given by $P^S = 0.2Q$, where Q is the quantity in tons and P is the price in thousands of francs per ton. The chemical

process in the production of fertilizer causes environmental damage of the order of 6 thousand francs per ton produced. This implies that the supply curve after taking into account social damage is $P^S = 0.2Q + 6$.

Which of the following statements is correct?

- (a) The socially optimal level of production is 70 and the deadweight loss is 75 francs.
- (b) The socially optimal level of production is 70 and the deadweight loss is 75,000 francs.
- (c) The socially optimal level of production is 5 tons higher than the unregulated level of production and the net loss is 150 francs.
- (d) The difference between the socially optimal level of production and the unregulated level of production is 5 tons and the net loss is 15,000 francs.

Exercise 12 (Exam January 2018)

Which of the following good satisfies both non-rivalry and exclusion (in consumption)? (a) Chocolate bar.

- (b) Garden with free access.
- (c) View of Mount Blanc. (d) Train (uncrowded).

Exercise 13 (Exam August 2018)

A production tax is imposed on a firm generating a negative externality to reach an economically efficient outcome. The optimal production tax is equal to:

- (a) the private marginal cost.
- (b) the social marginal cost.
- (c) the difference between the social marginal cost and the private marginal cost. (d) the sum of the social marginal cost and the private marginal cost.

Exercise 14

A risk-averse investor prefers to take a lower risk even if it means that he will have: (a) A lower return.

- (b) A better return.
- (c) A higher price.
- (d) A lower price.

Exercise 15 (Exam August 2017)

Which of the following propositions is correct?

- (a) The risk premium of a "risk-taker" individual is strictly positive.
- (b) Diversification of risks is an efficient method to reduce the risk of a single event only if the insurance policy is fair.
- (c) A "risk-taker" individual never refuses to participate in a lottery. (d) According to the "Prospect Theory", people tend to underestimate high probabilities.